

Module - 4Ques 1

#include <stdio.h>

int main()

{

long

 unsigned long int n;
 int count = 0;

 printf("Enter a number: ");
 scanf("%ld", &n);

while (n > 0)

{

count++;

n = n/10;

}

printf("Total number of digits = %d", count);

return 0;

{

Ques 2

#include <stdio.h>

int main() { long

unsigned long int n, rev = 0;

printf("Enter a no. : ");

scanf("%ld", &n);

while (n > 0)

{

rev = rev * 10 + (n % 10);

n = n/10;

}

```
printf ("Reversed number = %d", rev);
return 0;
}
```

Ques 3.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    unsigned long int n;
    int sum = 0;
```

```
printf ("Enter a no. : ");
scanf ("%ld", &n);
```

```
while (n > 0)
{
```

```
    sum = sum + (n % 10);
    n = n / 10;
```

```
}
```

```
printf ("Sum of digits = %d", sum);
```

```
return 0;
```

```
}
```


Ques4.

#include <stdio.h>

int main()

{

unsigned long long int n;

int even=0, odd=0, digit;

printf("Enter a no. : ");

scanf("%llu", &n);

while (n > 0)

{

digit = n % 10;

if (digit % 2 == 0)

even++;

else

odd++;

n = n / 10;

}

printf("Even digits = %d \n", even);

printf("Odd digits = %d \n", odd);

return 0;

{

Ques 5.

```
#include <stdio.h>
int main()
```

```
{
```

```
    unsigned long long int n;
    int largest = 0, digit;
```

```
    printf("Enter a number: ");
    scanf("%llu", &n);
```

```
    while (n > 0)
```

```
    {
```

```
        digit = n % 10;
```

```
        if (digit > largest)
            largest = digit;
```

```
        n = n / 10;
```

```
    }
```

```
    printf("Largest digit = %d", largest);
```

```
    return 0;
```

```
}
```


Ques. 6

#include <stdio.h>

int main()

{

int l, r, j, i, count = 0, prime;

printf("Enter starting & ending nos. : ");
scanf("%d %d", &l, &r);for (i = l; i <= r; i++)
{

if (i < 2)

continue;

prime = 1;

for (j = 2; j * j <= i; j++)
{if (i % j == 0)
{

prime = 0;

break;

}

}

if (prime)

count++;

}

printf("Total prime nos. = %d", count);

return 0;

{

Ques. 7

#include <stdio.h>

int main()

{

int n, i, sum = 0;

printf("Enter a no. : ");

scanf("%d", &n);

for (i = 1; i <= n/2; i++)

{

if (n % i == 0)

sum = sum + i;

}

printf("Sum of proper divisors = %d", sum);

return 0;

}

Ques. 8

#include <stdio.h>

int main()

{

int n, i;

printf("Enter a no. : ");

scanf("%d", &n);

printf("Factors are: ");

for (i = 1; i <= n; i++)

{

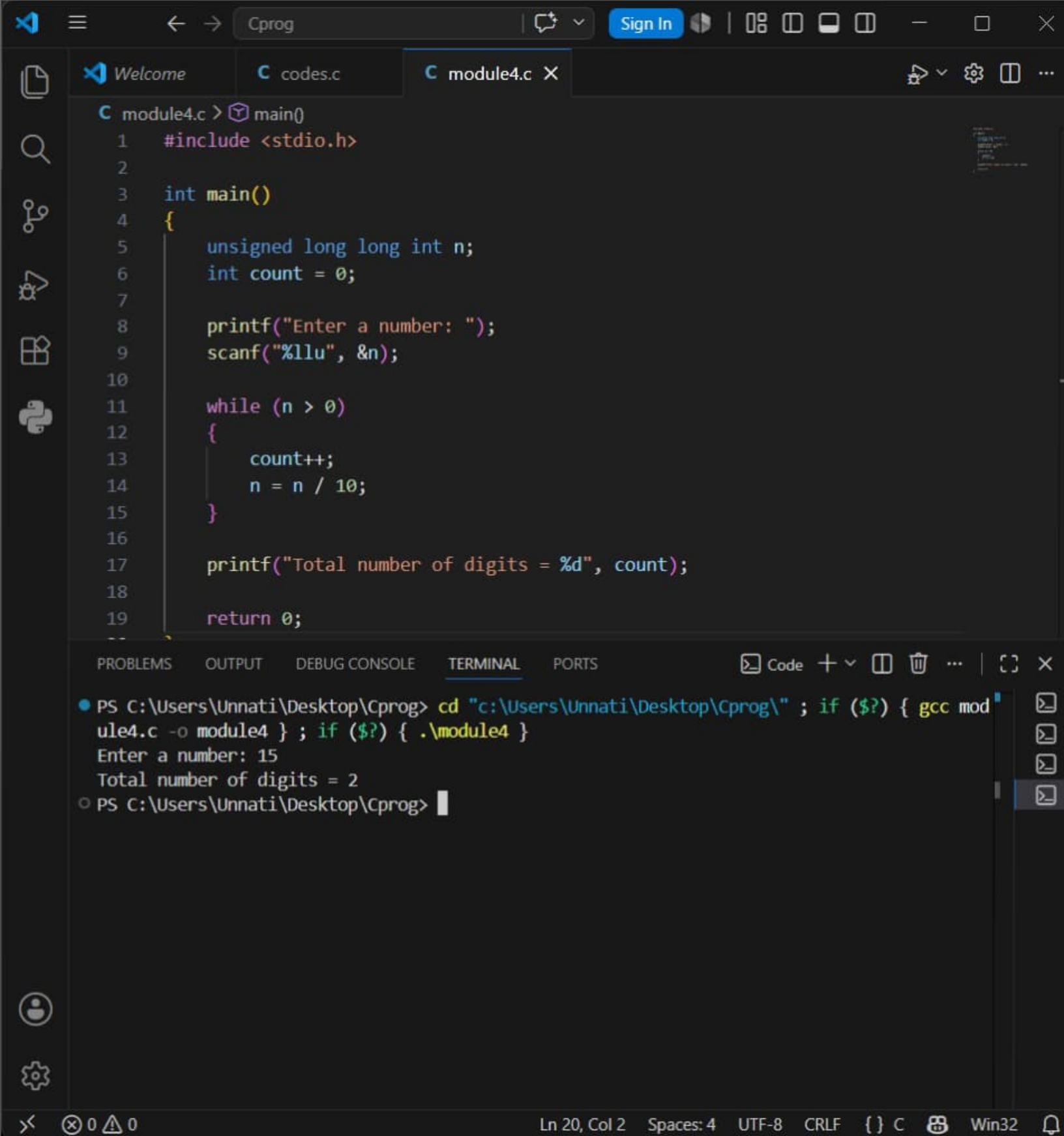
if (n % i == 0)

printf("%d ", i);

}

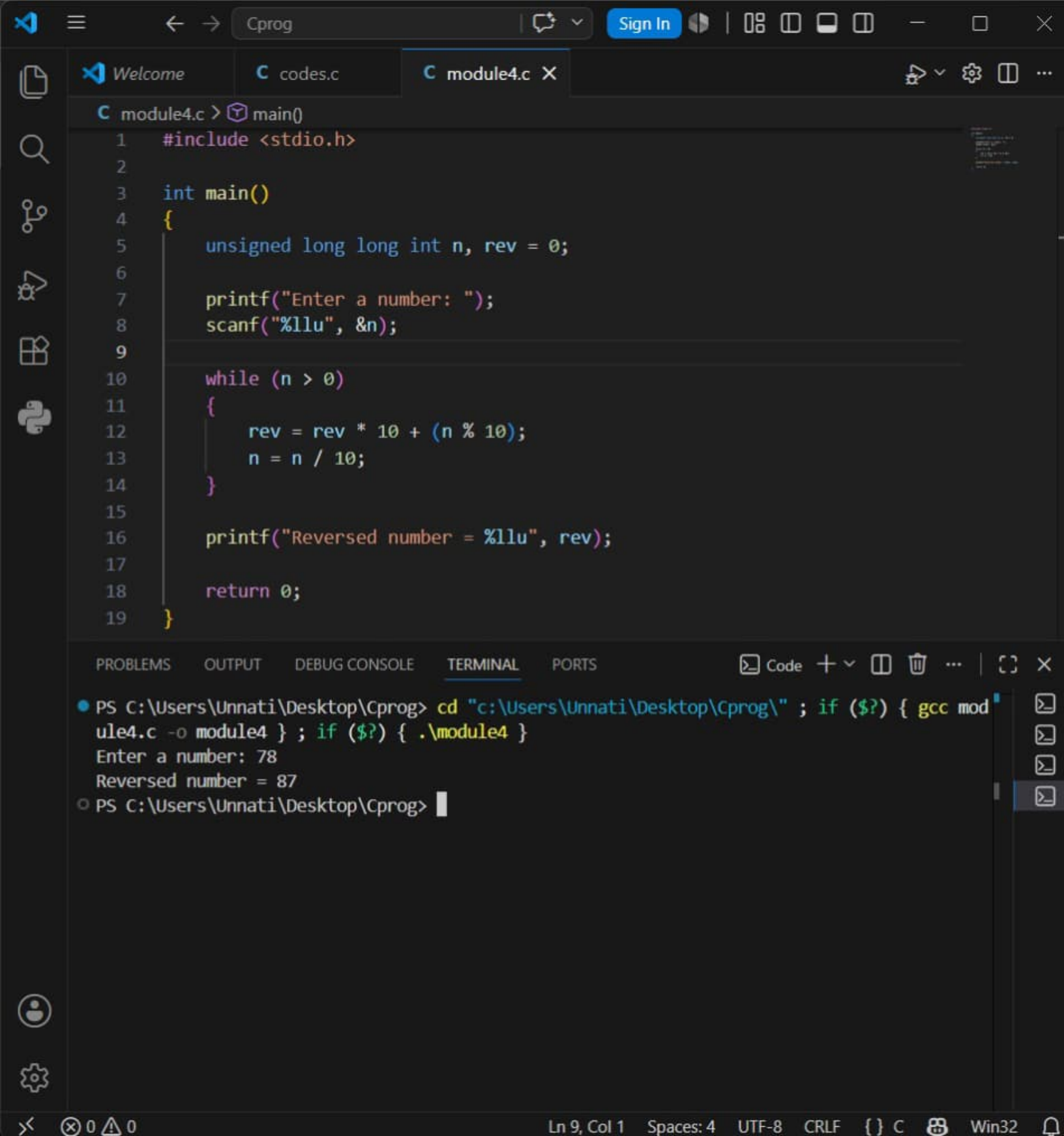
return 0;

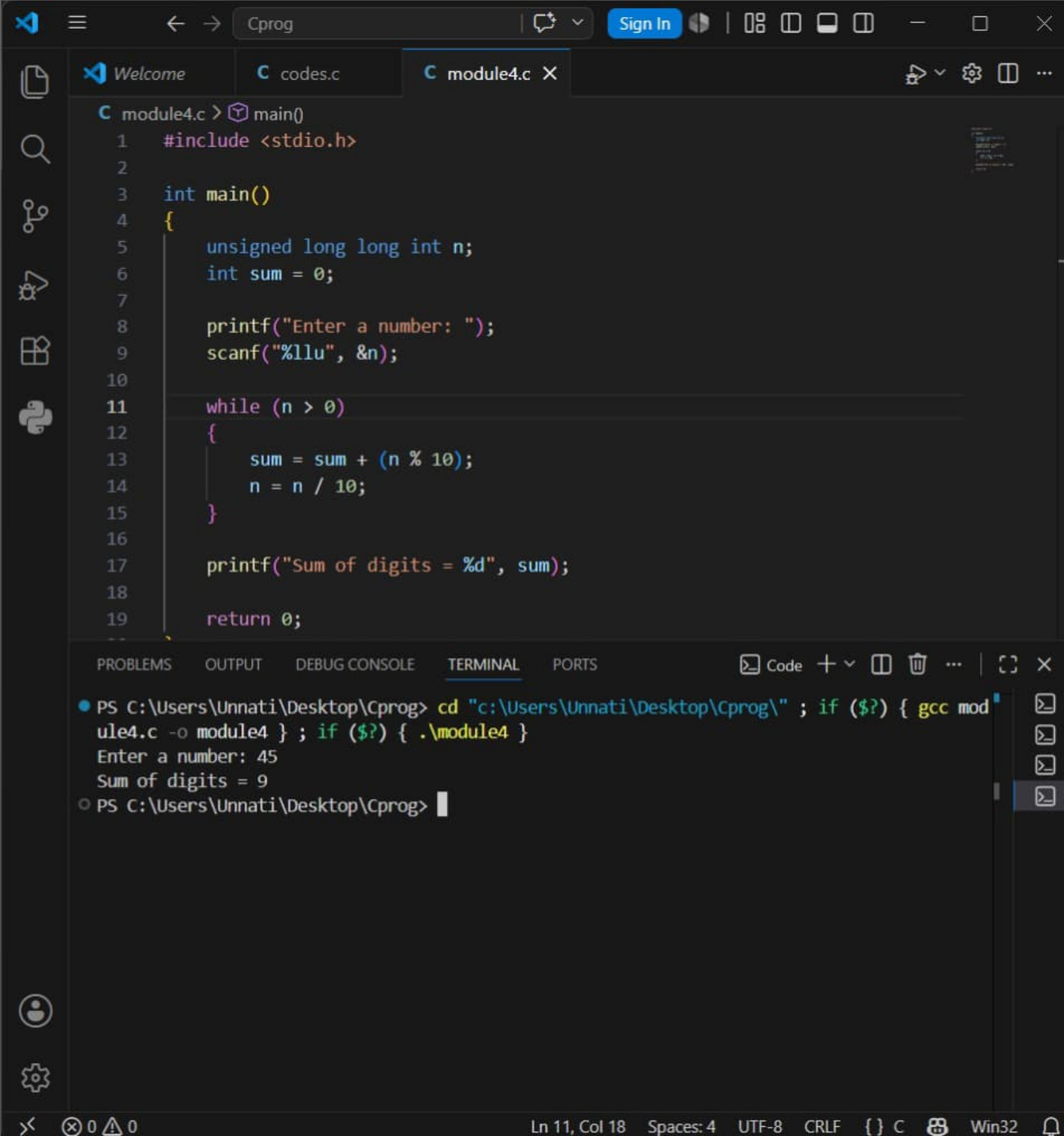
}

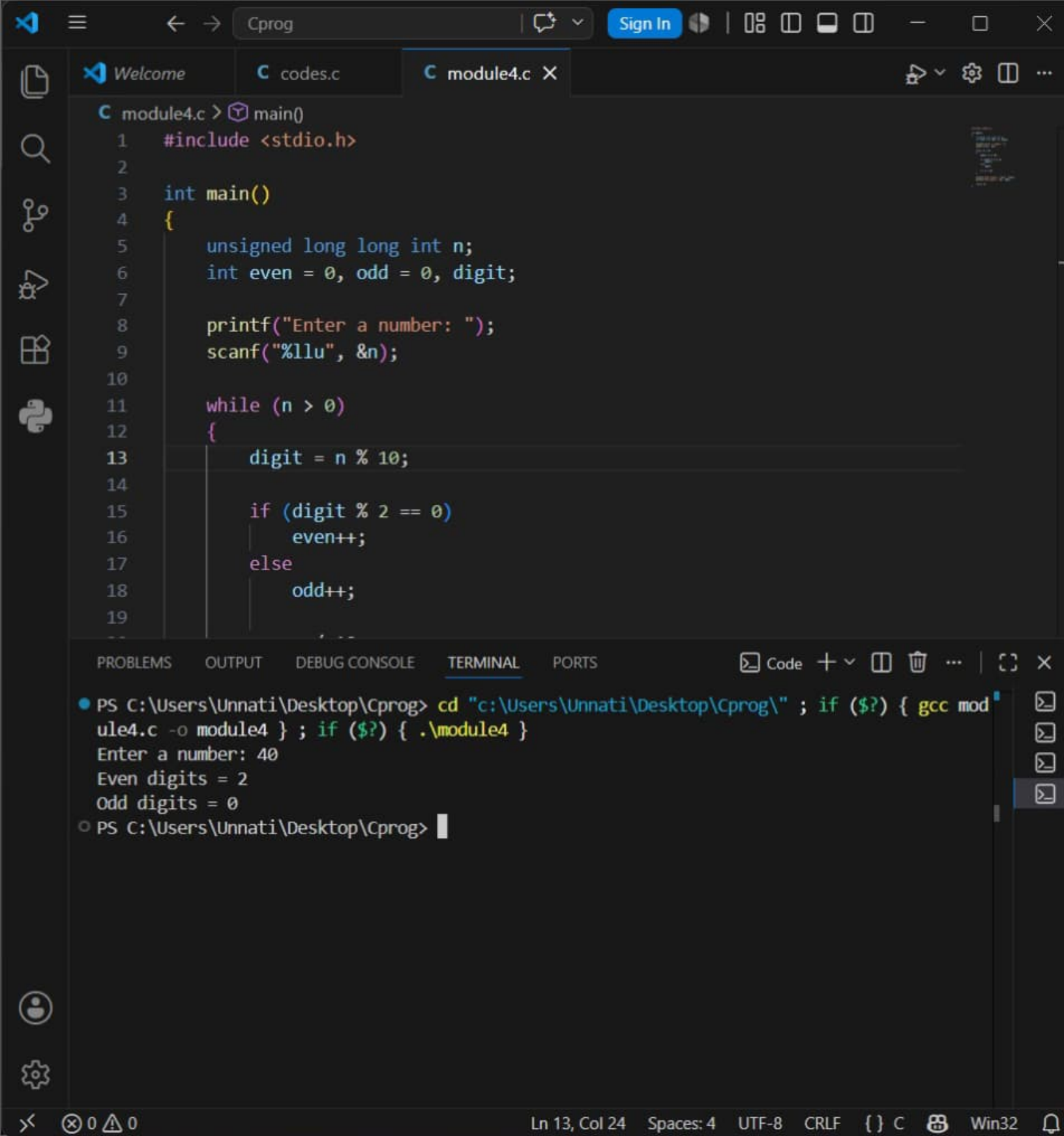


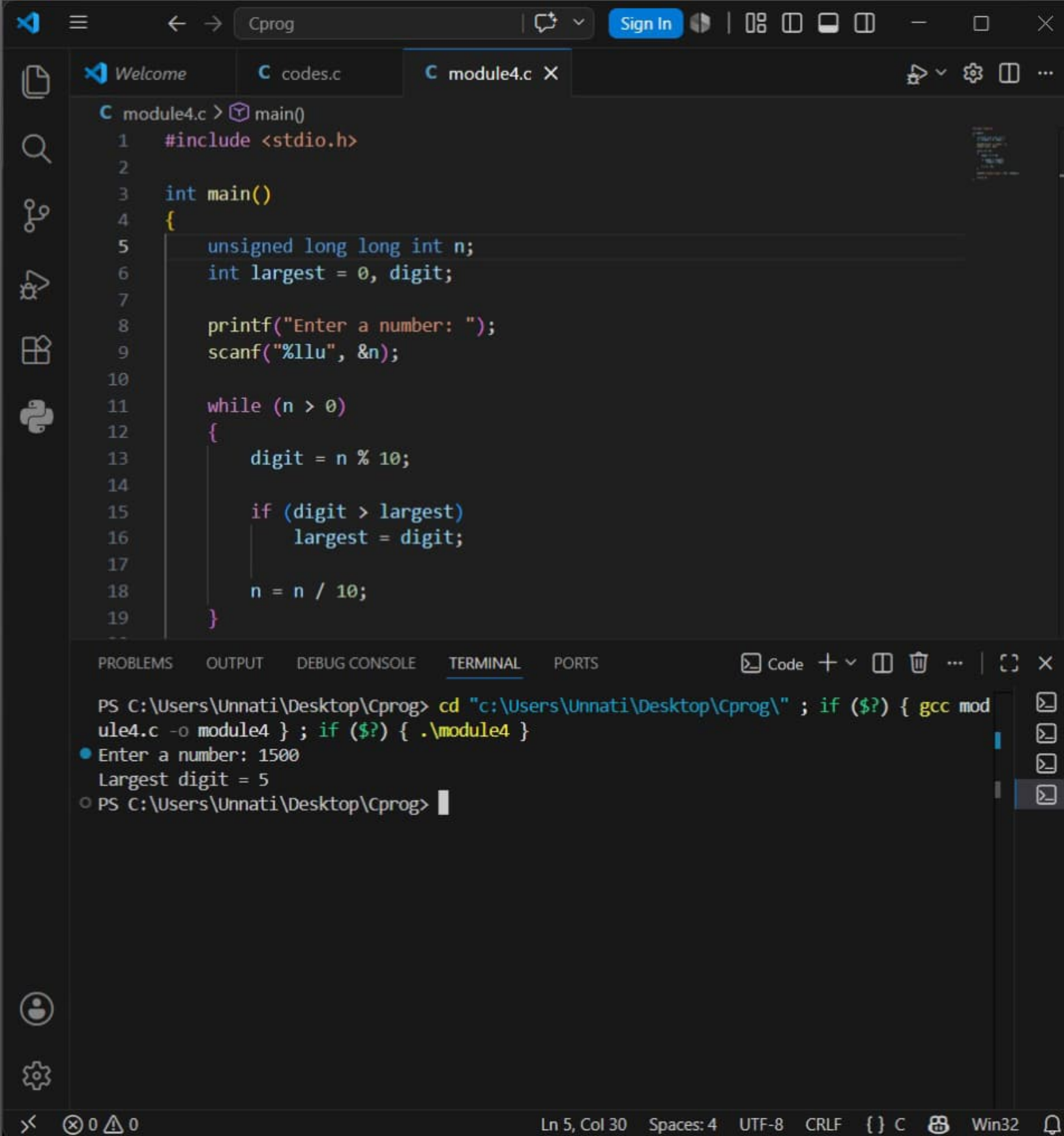
```
C module4.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      unsigned long long int n;
6      int count = 0;
7
8      printf("Enter a number: ");
9      scanf("%llu", &n);
10
11     while (n > 0)
12     {
13         count++;
14         n = n / 10;
15     }
16
17     printf("Total number of digits = %d", count);
18
19     return 0;
```

```
PS C:\Users\Unnati\Desktop\Cprog> cd "c:\Users\Unnati\Desktop\Cprog\" ; if ($?) { gcc module4.c -o module4 } ; if ($?) { .\module4 }
Enter a number: 15
Total number of digits = 2
PS C:\Users\Unnati\Desktop\Cprog>
```

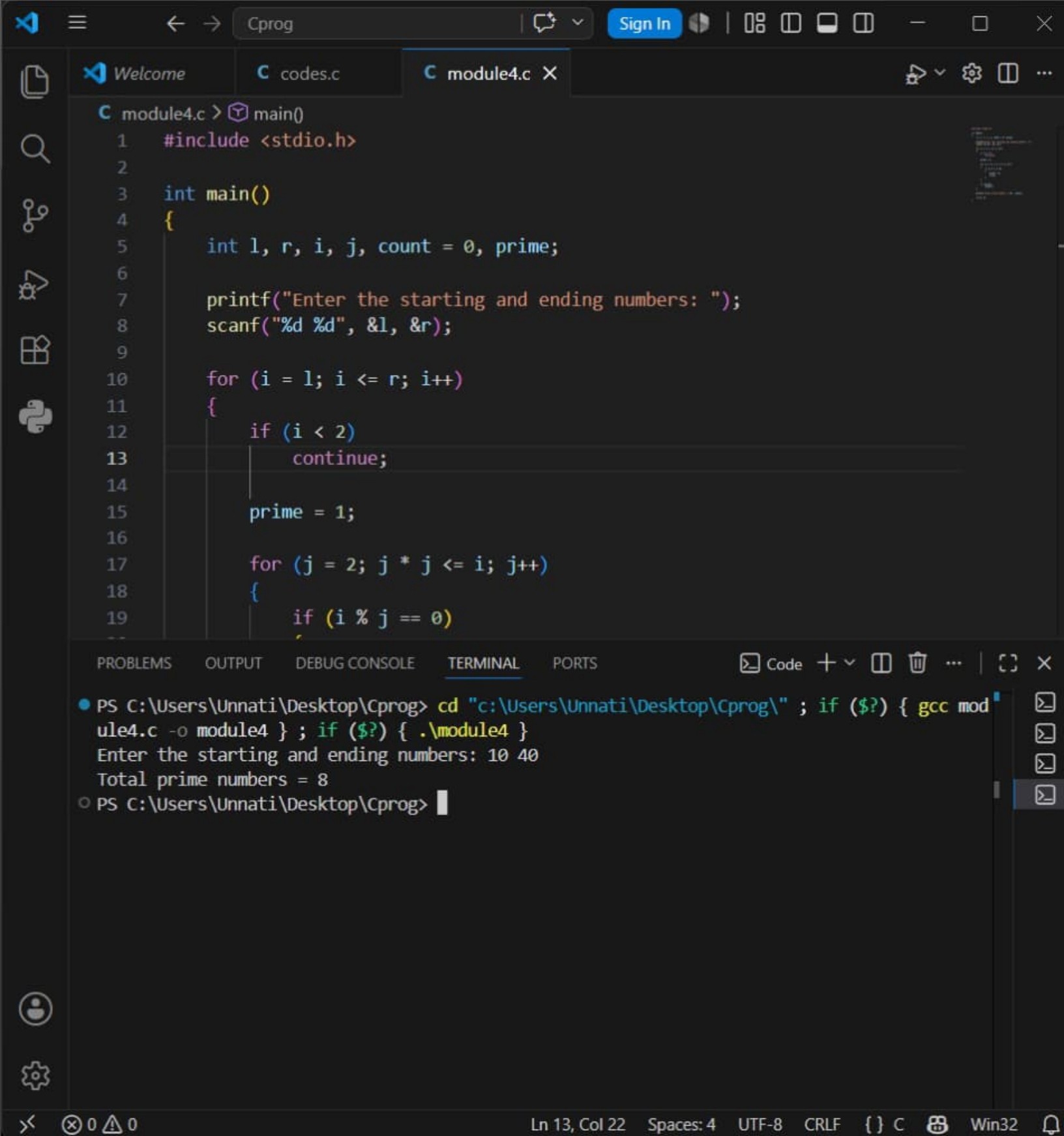






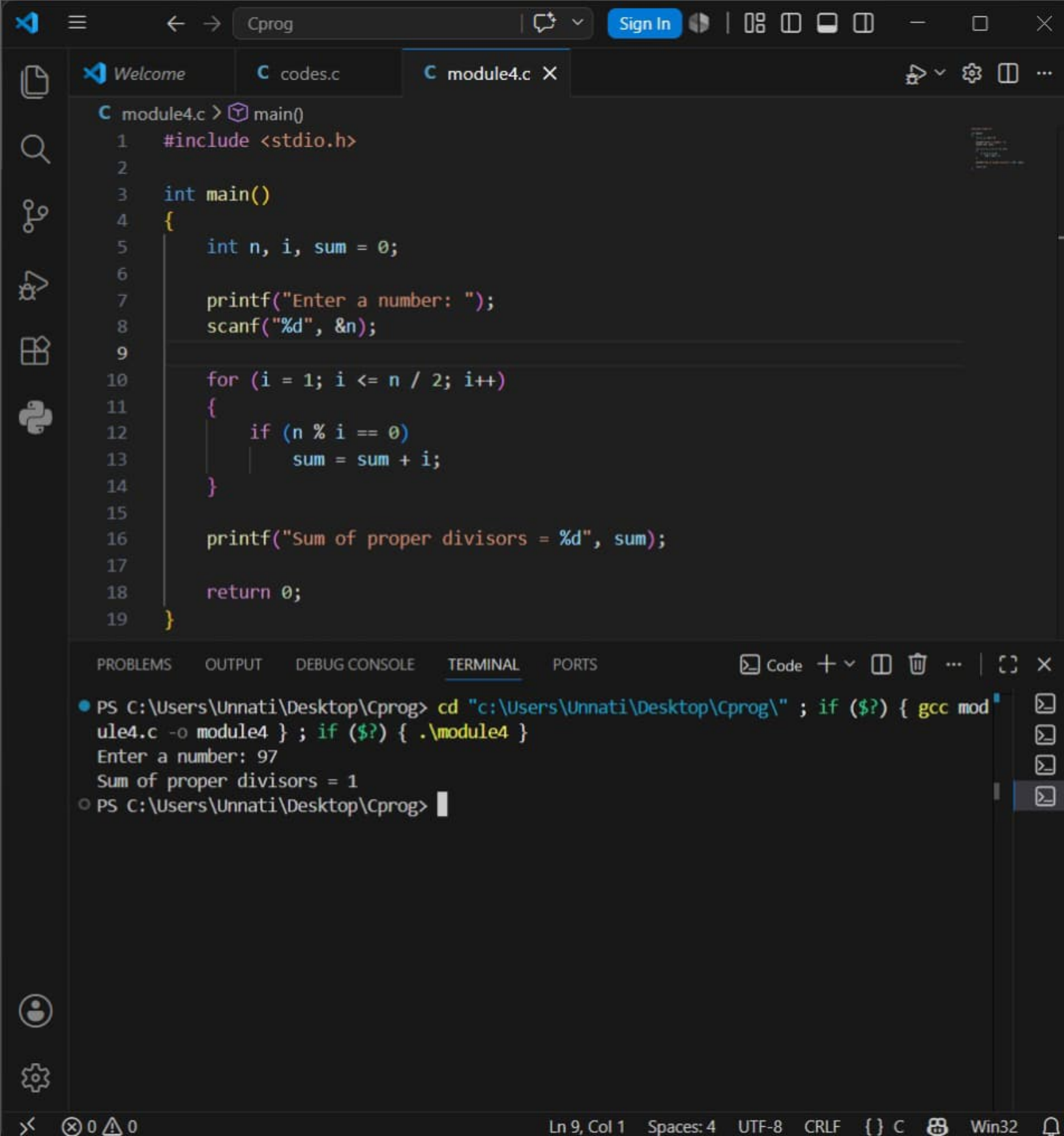


```
C module4.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      unsigned long long int n;
6      int largest = 0, digit;
7
8      printf("Enter a number: ");
9      scanf("%llu", &n);
10
11     while (n > 0)
12     {
13         digit = n % 10;
14
15         if (digit > largest)
16             largest = digit;
17
18         n = n / 10;
19     }
```



```
C module4.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      int l, r, i, j, count = 0, prime;
6
7      printf("Enter the starting and ending numbers: ");
8      scanf("%d %d", &l, &r);
9
10     for (i = l; i <= r; i++)
11     {
12         if (i < 2)
13             continue;
14
15         prime = 1;
16
17         for (j = 2; j * j <= i; j++)
18         {
19             if (i % j == 0)
```

```
PS C:\Users\Unnati\Desktop\Cprog> cd "c:\Users\Unnati\Desktop\Cprog\" ; if ($?) { gcc mod
ule4.c -o module4 } ; if ($?) { .\module4 }
Enter the starting and ending numbers: 10 40
Total prime numbers = 8
PS C:\Users\Unnati\Desktop\Cprog>
```

```
C module4.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i;
6
7      printf("Enter a number: ");
8      scanf("%d", &n);
9
10     printf("Factors are: ");
11
12     for (i = 1; i <= n; i++)
13     {
14         if (n % i == 0)
15             printf("%d ", i);
16     }
17
18     return 0;
19 }
```

```
PS C:\Users\Unnati\Desktop\Cprog> cd "c:\Users\Unnati\Desktop\Cprog\" ; if ($?) { gcc module4.c -o module4 } ; if ($?) { .\module4 }
Enter a number: 56
Factors are: 1 2 4 7 8 14 28 56
PS C:\Users\Unnati\Desktop\Cprog>
```